PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

EDU - INFO - PORTAL

A PROJECT REPORT

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IN

COMPUTER SCIENCE AND ENGINEERING

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this **report "Disaster preparedness and response education system for schools and colleges"** is a Bonafide work of **"Aadhish Pranav P D (20221CSE0095), Amith H S (20221CSE0113), Uday Kumar Reddy (20221CSE0241)"**, who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING** during 2025-26.

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DECLARATION

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Abstract

Educational information such as government policies, academic regulations, scholarship schemes, higher study opportunities, and examination notifications is often distributed across multiple platforms, making it difficult for users to access relevant and reliable information efficiently. This lack of centralization leads to increased search time, confusion, and missed opportunities for students.

This project presents the **Edu Info Portal**, a centralized web-based platform designed to organize and provide easy access to educational information. The system is developed using a full-stack architecture consisting of a React.js frontend, Node.js and Express.js backend, and MongoDB Atlas cloud database.

The portal categorizes information into multiple modules such as policies, schemes, regulations, scholarships, higher studies, and educational news. The system uses RESTful APIs to dynamically retrieve data from the database and display it through an interactive user interface. Additional features include a **login and registration system**, **smart search functionality**, and a **chatbot interface** that assists users in quickly accessing relevant information. The chatbot uses keyword-based detection to guide users to appropriate sections of the portal.

The system is deployed using cloud platforms, with the frontend hosted on Vercel and the backend on Render, ensuring accessibility and scalability. The implementation demonstrates the effectiveness of centralized information systems in improving accessibility, reducing search effort, and enhancing user experience.

Future enhancements may include AI-based chatbot integration, advanced filtering mechanisms, and secure authentication systems to further improve functionality.

Table of Content

Sl. No.	Title	Page No.
	Declaration	iii
	Acknowledgement	iv
	Abstract	v
	List of Figures	viii
	List of Tables	ix
	Abbreviations	x
1.	Introduction 1.1 Background 1.2 Statistics 1.3 Prior existing technologies 1.4 Proposed approach 1.5 Objectives 1.6 SDGs 1.7 Overview of project report	1-6
2.	Literature review	7-11
3.	Methodology	12-16
4.	Project management 4.1 Project timeline 4.2 Risk analysis 4.3 Project budget	17-19
5.	Analysis and Design 5.1 Requirements 5.2 System Architecture 5.3 Data Flow Diagram 5.4 System Flow Chart 5.5 Module Design 5.6 Database Design 5.7 API Design	20-34

	5.8 UI/UX Design 5.9 Technology Stack Justification 5.10 Security & Performance 5.11 Deployment Architecture 5.12 Constraints & Assumptions	
6.	Hardware, Software and Simulation 6.1 Hardware 6.2 Software development tools 6.3 Software code 6.4 Simulation	35-37
7.	Evaluation and Results 7.1 Test points 7.2 Test plan 7.3 Test result 7.4 Insights	38-42
8.	Social, Legal, Ethical, Sustainability and Safety Aspects 8.1 Social aspects 8.2 Legal aspects 8.3 Ethical aspects 8.4 Sustainability aspects 8.5 Safety aspects	43-45
9.	Conclusion	46-48
	References	49
	Base Paper	51
	Appendix	52-61

List of Figures

Figure	Caption	Page no
Fig 1.1	Sustainable development goals	5
Fig 3.1	System Architecture – MVC Layered Model	13
Fig 5.1	System Architecture	22
Fig 5.2	Data Flow Diagram	24
Fig 5.3	Flow Chart	25
Fig 5.4	Architecture of Deployment	32
Fig A	GitHub Repository	50
Fig B	Website Login Page	51
Fig C	Register Page	51
Fig D	Home Page	52
Fig E	Schemes Dashboard	52
Fig F	Regulation Dashboard	53
Fig G	News Dashboard	53
Fig H	Chatbot Response	54
Fig I	Chatbot Response	55
Fig J	Database Mongo DB Atlas	56
Fig K	MongoDB Atlas – Policies Collection Data	56
Fig L	MongoDB Atlas – Scholarships Collection Data	57
Fig M	MongoDB Atlas – Schemes Collection Data	57
Fig N	Similarity Report	58
Fig O	AI Similarity Report	58
Fig P	Submission confirmation email for ICACECS 2026 paper	59

List of Tables

Table	Caption	Page no
Table 1.1	Technologies that existed earlier	2
Table 1.2	Objectives categorized by system aspects	5
Table 2.1	Summary of Literature Reviews	9-11
Table 4.1	Project Timeline	18
Table 4.2	Risk Analysis	18
Table 4.3	Project Budget Estimation	19
Table 5.1	Functional Requirements	27
Table 5.2	Non- Functional Requirements	28
Table 5.3	Module Overview	29
Table 5.4	Database Collections and Fields	28
Table 6.1	Hardware Requirements	34
Table 6.2	Software Development Tools	35
Table 7.1	Test Points	37
Table 7.2	Test Plan	38
Table 7.3	Test Results Summary	40

Abbreviations

S.no	Abbreviation	Full Form
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	CSS	Cascading Style Sheets
4	DFD	Data Flow Diagram
5	SDG	Sustainable Development Goals
6	GSAP	GreenSock Animation Platform
7	HTTP	HyperText Transfer Protocol
8	HTML	HyperText Markup Language
9	IoT	Internet of Things
10	IoTWF	Internet of Things World Forum
11	JSON	JavaScript Object Notation
12	LLM	Large Language Model
13	NFR	Non-Functional Requirement
14	FR	Functional Requirement
15	UI	User Interface
16	UX	User Experience
17	CPU	Central Processing Unit
18	RAM	Random Access Memory
19	GPU	Graphics Processing Unit
20	OS	Operating System
21	DB	Database
22	URL	Uniform Resource Locator
23	REST	Representational State Transfer
24	CRUD	Create, Read, Update, Delete

Chapter 1

Introduction

1.1 Background

In today's rapidly evolving educational environment, students and learners are required to stay updated with a wide range of information such as government policies, scholarship schemes, academic regulations, higher education opportunities, and examination updates. However, this information is widely scattered across multiple platforms including government websites, university portals, and third-party resources.

The absence of a unified platform makes it difficult for users to access relevant and reliable information efficiently. Students often spend significant time searching across different sources, which may lead to confusion or missed opportunities. In many cases, lack of awareness prevents students from benefiting from available opportunities such as scholarships and government schemes.

To address these challenges, the **Edu Info Portal** is developed as a centralized web-based system that integrates multiple categories of educational information into a single platform. The system is designed using modern web technologies to provide a structured, accessible, and user-friendly experience. By organizing data into clearly defined categories and enabling dynamic retrieval, the platform simplifies the process of accessing educational resources.

1.2 Statistics

The need for a centralized educational information system is further supported by the following observations:

- India has over **800 million internet users**, making digital platforms highly accessible.
- A large number of students rely on **multiple websites** to gather educational information.
- Many students **miss scholarship and scheme opportunities** due to lack of awareness.
- Educational data available online is often **unstructured and difficult to navigate**.
- Users spend considerable time filtering irrelevant results from search engines.
- Centralized platforms help in **reducing search time and improving accessibility**.

These observations highlight the importance of developing a unified system that can efficiently organize and present educational information.

1.3 Prior Existing Technologies

Before the development of centralized platforms like Edu Info Portal, users depended on various technologies to access educational information:

Table 1.1: Technologies that existed earlier

Technology	Contribution	Limitations
Government Websites	Provide official and authentic data	Complex navigation and scattered information
Institutional Portals	Offer academic-specific updates	Limited to individual institutions
Search Engines	Easy access to multiple sources	Requires manual filtering of results
Social Media Platforms	Quick dissemination of updates	Lack of reliability and structure
Static Websites	Provide informational content	No dynamic interaction or personalization

Limitations of Existing Technologies

- Lack of centralized access to information
- Absence of structured categorization
- No intelligent or guided search system
- Limited user interaction
- Time-consuming manual searching

These limitations emphasize the need for a more efficient and integrated system.

1.4 Proposed Approach

To overcome the limitations of existing systems, the Edu Info Portal proposes a centralized and interactive solution that enables users to access educational information easily and efficiently.

1.4.1 Aim of the Project

The primary aim of the project is to develop a **centralized web-based platform** that provides structured and easily accessible educational information, improving user experience and reducing search effort.

1.4.2 Proposed Approach

The Edu Info Portal is implemented using the MERN stack architecture:

- **Frontend:** React.js with Bootstrap for responsive UI
- **Backend:** Node.js with Express.js for API handling
- **Database:** MongoDB Atlas for cloud-based data storage

The system organizes data into the following modules:

- Policies
- Schemes
- Regulations
- Scholarships
- Higher Studies
- Educational News

The platform uses RESTful APIs to dynamically retrieve and display data from the database, ensuring scalability and flexibility.

1.4.3 Key Features

The system integrates multiple modern features to enhance usability and accessibility:

- Centralized educational information system
- Dynamic routing with individual detail pages

- Login and Registration system for user authentication
- Logout functionality with session handling
- Chatbot integration for quick information access
- Smart search functionality across multiple categories
- Responsive and user-friendly interface
- Cloud database integration using MongoDB Atlas
- Deployment using Vercel (frontend) and Render (backend)

1.4.4 Limitations of the Proposed Approach

Despite its advantages, the system has certain limitations:

- Chatbot is currently **keyword-based**, not AI/NLP-based
- Authentication uses **local storage**, which is not fully secure
- Data updates depend on **manual database entry**
- No real-time notification system
- Limited personalization features

1.5 Objectives

The objectives of the Edu Info Portal are:

1. To develop a centralized platform for educational information
2. To enable dynamic data retrieval using REST APIs
3. To design a responsive and user-friendly interface
4. To implement a scalable and modular system architecture
5. To integrate chatbot-based assistance for users
6. To provide basic user authentication functionality
7. To deploy the system on cloud platforms for global accessibility

1.6 Sustainable Development Goals (SDGs)

Table 1.2 : Objectives categorized by system aspects

SDG	Goal	Contribution
SDG 4	Quality Education	Improves access to educational resources
SDG 9	Industry, Innovation & Infrastructure	Utilizes modern web technologies
SDG 10	Reduced Inequalities	Provides equal access to opportunities
SDG 16	Strong Institutions	Enhances transparency of information

Fig 1.1 Sustainable Development Goals

1.7 Overview of the Project Report

This report is organized to systematically present the design, implementation, and evaluation of the Edu Info Portal:

- **Chapter 1:** Introduction, background, and objectives
- **Chapter 2:** Literature review of existing systems
- **Chapter 3:** Methodology and system design
- **Chapter 4:** Project management (timeline, risk, budget)
- **Chapter 5:** Analysis and design (architecture, UI, APIs, database)
- **Chapter 6:** Hardware, software, and implementation
- **Chapter 7:** Evaluation and results
- **Chapter 8:** Social, legal, ethical, and sustainability aspects
- **Chapter 9:** Conclusion and future scope

Chapter 2

Literature Review

The development of centralized educational information systems has been influenced by advancements in web technologies, information systems, and digital learning platforms. Several studies have highlighted the importance of structured data access, scalable architectures, and user-friendly interfaces in improving accessibility to educational resources.

Kitchin [1] discusses the concept of real-time data systems and emphasizes the importance of centralized platforms in managing large-scale information efficiently. This concept is relevant to the Edu Info Portal, where educational data is collected and presented through a unified interface.

Anthopoulos [2] highlights the role of digital governance and smart information systems in improving accessibility and transparency. The study suggests that centralized digital platforms can significantly enhance the delivery of public information services, which aligns with the objective of providing structured educational data in this project.

Erl [3] introduces the concept of Service-Oriented Architecture (SOA), which focuses on modular and reusable services. Similarly, the Edu Info Portal adopts a modular approach using RESTful APIs, allowing efficient communication between the frontend and backend.

Fowler [4] discusses enterprise application architecture patterns, emphasizing scalability and maintainability. These principles are applied in the project through the use of a three-tier architecture consisting of frontend, backend, and database layers.

Satzinger et al. [5] and Alter [6] provide insights into system analysis and information systems design. Their work highlights the importance of structured system development methodologies, which have been followed in designing the Edu Info Portal.

Cloud-based architectures discussed by Amazon Web Services [7] and Google Cloud [8] demonstrate how scalable web applications can be built using distributed systems. The use of MongoDB Atlas and cloud deployment platforms such as Render and Vercel in this project reflects these practices.

Banks and Porcello [9] emphasize modern frontend development using React, focusing on component-based architecture and dynamic user interfaces. This approach is used in the project to create reusable UI components and improve user experience.

Cantelon et al. [10] describe backend development using Node.js, highlighting its efficiency in handling asynchronous operations and API development. The backend of the Edu Info Portal uses Node.js and Express.js to manage data retrieval and routing.

Alammary [11] and Ally & Khan [12] discuss the role of digital learning systems and e-learning platforms in modern education. Their studies emphasize the need for accessible and interactive systems, which is addressed through features like chatbot interaction and smart search.

Gupta and Bansal [13] highlight the importance of web-based educational systems in improving student services by providing centralized access to information. Similarly, Kumar and Singh [14] emphasize the effectiveness of centralized portals in enhancing accessibility and reducing information fragmentation.

Chen et al. [15] discuss scalable web architectures for information retrieval systems, focusing on performance and efficiency. These principles are reflected in the project through the use of REST APIs and optimized data retrieval mechanisms.

Summary

The reviewed literature highlights the importance of:

- Centralized information systems for efficient data access
- Scalable and modular web architectures
- Cloud-based data storage solutions
- User-friendly interfaces for improved accessibility

- Integration of modern technologies such as React and Node.js

The Edu Info Portal builds upon these concepts by integrating a centralized database, modular architecture, chatbot interaction, and smart search functionality to provide a comprehensive educational information system.

Table 2.1: Summary of Literature Reviews

Reference	**Focus Area**	**Key Findings**	**Accuracy / Performance**	**Limitations**
Kitchin [1]	Real-time data systems	Centralized platforms improve large-scale data management	Efficient handling of dynamic data	Requires high computational resources
Anthopoulos [2]	Digital governance & smart systems	Enhances accessibility and transparency of information	Improves public service delivery	Implementation complexity
Erl [3]	Service-Oriented Architecture (SOA)	Promotes modular and reusable system components	High flexibility and scalability	Increased system design complexity
Fowler [4]	Enterprise architecture patterns	Supports scalable and maintainable application design	Improves system reliability	Requires careful architectural planning
Satzinger et al. [5]	System analysis & design	Structured methodologies improve system development	Enhances system organization	Time-consuming development process

Alter [6]	Information systems design	Emphasizes structured system workflows	Improves efficiency and clarity	Limited flexibility in dynamic environments
AWS [7]	Cloud computing	Enables scalable and distributed applications	High availability and scalability	Dependency on internet connectivity
Google Cloud [8]	Distributed systems	Supports performance optimization and scalability	Efficient data processing	Cost management challenges
Banks & Porcello [9]	Frontend development (React)	Component-based UI improves reusability	Enhances user experience	Learning curve for beginners
Cantelon et al. [10]	Backend development (Node.js)	Efficient handling of asynchronous operations	High performance in real-time apps	Callback complexity
Alammary [11]	Digital learning systems	Improves accessibility in education	Supports interactive learning	Limited personalization
Ally & Khan [12]	E-learning platforms	Encourages interactive and flexible learning	Enhances engagement	Requires stable internet access
Gupta & Bansal [13]	Web-based education systems	Centralized access improves student services	Reduces information gaps	Limited real-time updates
Kumar & Singh [14]	Centralized portals	Reduces data fragmentation	Improves accessibility	Integration challenges

Chen et al. [15]	Scalable web systems	Optimizes data retrieval and system performance	High efficiency and speed	Requires optimization strategies

Chapter 3

Methodology

The development of the Edu Info Portal follows a structured methodology that integrates system design, data management, and user interaction. The system is built using a full-stack approach to ensure efficient data handling, scalability, and a seamless user experience. The methodology focuses on designing a centralized platform that allows users to access educational information through categorized modules, smart search functionality, and chatbot interaction.

3.1 System Development Approach

The project follows a **modular and incremental development approach**, where different components of the system are developed and integrated step by step.

The development process includes:

1. **Requirement Analysis**
   Identification of user needs such as accessing policies, schemes, scholarships, and educational updates.
2. **System Design**
   Designing frontend, backend, and database architecture.
3. **Implementation**
   Developing modules using React, Node.js, and MongoDB.
4. **Integration**
   Connecting frontend with backend APIs and database.
5. **Testing**
   Verifying functionality, performance, and user interaction.
6. **Deployment**
   Hosting frontend and backend on cloud platforms.

3.2 System Architecture

The Edu Info Portal follows a **three-tier architecture**, which separates the system into different layers:

1. Presentation Layer (Frontend)

- Developed using React.js
- Provides user interface and interaction
- Handles navigation, search, and display of data

2. Application Layer (Backend)

- Developed using Node.js and Express.js
- Handles API requests and business logic
- Processes user queries and retrieves data

3. Data Layer (Database)

- Implemented using MongoDB Atlas
- Stores structured educational data
- Supports efficient querying and scalability

This architecture ensures separation of concerns and improves maintainability.

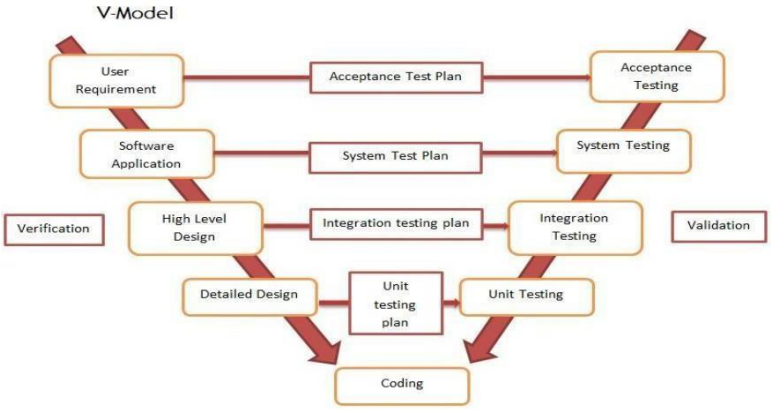


Fig 3.1: System Architecture – MVC Layered Model

3.3 Data Flow Methodology

The system follows a structured data flow process:

1. User interacts with the frontend interface
2. User performs actions such as search or chatbot query
3. Frontend sends API request to backend
4. Backend processes request using Express routes
5. Backend queries MongoDB database
6. Data is retrieved and sent back to frontend
7. Frontend dynamically displays results

This process ensures efficient communication between system components.

3.4 Functional Workflow

The functional workflow of the system includes:

User Authentication Workflow

- User registers using email and password
- Credentials are stored in local storage
- User logs in and session is maintained
- Logout clears session data

Search Workflow

- User enters query in search bar
- Request is sent to backend /api/search
- Backend searches across multiple collections
- Matching results are returned and displayed

Chatbot Workflow

- User enters query in chatbot
- Keyword-based detection identifies category
- Backend API is called based on keyword
- Results are displayed as interactive cards

3.5 Smart Search

This The smart search functionality is implemented using:

- Keyword-based matching
- Regular expressions for flexible search
- Multi-collection querying
- Dynamic routing to detail pages

This enables users to directly access relevant information without manual navigation.

3.6 Chatbot

The chatbot is designed as a lightweight assistance system that improves user interaction.

Features include:

- Keyword detection (news, schemes, policies, etc.)
- Dynamic API calls
- Interactive result display
- Quick navigation to detailed pages

Although not fully AI-based, the chatbot enhances accessibility and usability.

3.7 Deployment

The system is deployed using cloud platforms:

- **Frontend:** Vercel
- **Backend:** Render
- **Database:** MongoDB Atlas

Deployment steps include:

1. Code pushed to GitHub
2. Backend deployed on Render
3. Frontend deployed on Vercel
4. Environment variables configured
5. APIs connected to live backend

This ensures global accessibility and scalability.

3.8 Advantages

- Modular and scalable architecture
- Efficient data retrieval using APIs
- Centralized data management
- Improved user interaction through chatbot
- Cloud-based deployment for accessibility

3.9 Limitations of the Methodology

- Keyword-based chatbot lacks AI intelligence
- Local storage authentication is not highly secure
- No real-time data synchronization
- Limited advanced filtering features

Chapter 4

Project Management

Project management plays a crucial role in ensuring the successful development and deployment of the Edu Info Portal. It involves planning, organizing, and managing resources effectively to complete the project within the given time and constraints. The project was executed in multiple phases, including requirement analysis, design, development, testing, and deployment.

4.1 Project Timeline

The development of the Edu Info Portal was carried out in a structured and phased manner aligned with scheduled project reviews. Each phase of the project focused on specific tasks, ensuring systematic progress from problem identification to final deployment.

The project began with proposal development and requirement analysis during Review-1. This was followed by system design and architecture planning, which was evaluated during Review-2. The initial implementation of the frontend and core functionalities was presented during Review-3.

Subsequently, advanced features such as backend integration, chatbot functionality, and database connectivity were developed and demonstrated during Review-4. Final enhancements, including the login system, smart search functionality, and deployment on cloud platforms, were completed before the final viva.

This structured timeline ensured efficient project execution, timely completion of milestones, and continuous evaluation at each stage.

Table 4.1: Project Timeline

Sl. No	Phase / Activity	Description	Start Date	End Date
1	Review-1 (Proposal)	Problem identification, idea finalization, initial planning	10-01-2026	16-01-2026
2	System Design Phase	Architecture design, module planning, UI structure	17-01-2026	30-01-2026
3	Review-2 (Design Review)	Presentation of system design and architecture	31-01-2026	06-02-2026
4	Development Phase – I	Frontend development (UI, components, navigation)	07-02-2026	27-02-2026
5	Review-3 (Implementation Review)	Initial implementation demonstration	28-02-2026	06-03-2026
6	Development Phase – II	Backend APIs, database integration, chatbot	07-03-2026	02-04-2026
7	Review-4 (Pre-Final Review)	Complete system demonstration with features	03-04-2026	09-04-2026
8	Final Enhancements	Login system, smart search, deployment fixes	10-04-2026	19-04-2026
9	Final Viva	Final project evaluation and submission	20-04-2026	26-04-2026

4.2 Risk Analysis

During the development of the project, several potential risks were identified and mitigated.

Table 4.2 : Risk Analysis

Risk	Description	Impact	Mitigation
API Failure	Backend APIs may fail or crash	High	Proper error handling and testing
Database Connectivity	MongoDB connection issues	High	Use of cloud database and retry logic
Deployment Errors	Issues during hosting on Render/Vercel	Medium	Manual deployment and logs monitoring
Authentication Issues	Login/logout malfunction	Medium	Local storage handling and validation

UI Bugs	Layout or responsiveness issues	Low	Cross-device testing
Search Failure	Incorrect or no results returned	Medium	Regex-based search optimization

4.3 Project Budget

The Edu Info Portal was developed using open-source technologies and free-tier cloud services, eliminating the need for significant financial investment. The project primarily utilized tools such as Visual Studio Code for development, GitHub for version control, and MongoDB Atlas for database management.

Deployment was carried out using Vercel for the frontend and Render for the backend, both of which provide free hosting options suitable for academic projects. The use of these technologies ensured cost-effective development while maintaining scalability and performance. Thus, the project demonstrates an efficient use of available resources without incurring additional financial costs.

Table 4.3: Project Budget Estimation

Resource Type	**Tool / Platform Used**	**Purpose**
Development Environment	Visual Studio Code	Code development
Version Control	GitHub	Code management & collaboration
Frontend Hosting	Vercel	Deployment of React application
Backend Hosting	Render	Deployment of Node.js server
Database	MongoDB Atlas	Cloud-based data storage
Programming Languages	JavaScript	Application development
Frameworks	React.js, Node.js, Express.js	Full-stack development
UI Design	Bootstrap, CSS	Responsive interface design
Testing	Browser DevTools	Debugging and testing

Chapter 5

Analysis and Design

The This chapter presents a detailed analysis and design of the Edu Info Portal system. It focuses on understanding system requirements, designing system architecture, defining data flow, and structuring modules to ensure efficient functionality. The system is designed to provide a scalable, user-friendly, and centralized platform for accessing educational information.The design approach emphasizes modularity, dynamic data handling, and seamless interaction between frontend, backend, and database components.

5.1 Requirements

System requirements define the functionalities and constraints of the application.

Functional Requirements

The system should:

- Allow users to register and login
- Enable users to browse categories such as policies, schemes, etc.
- Display detailed information for each item
- Provide a smart search feature
- Support chatbot-based query handling
- Retrieve data dynamically from the database
- Allow navigation between different modules

Non-Functional Requirements

The system should:

- Provide a responsive user interface
- Ensure fast response time
- Support scalability for future expansion
- Maintain data consistency
- Be accessible through web browsers
- Ensure basic data security

5.2 System Architecture

The Edu Info Portal is designed using a **three-tier architecture**, which divides the system into distinct layers to ensure modularity, scalability, and maintainability. This architectural approach helps in separating concerns and allows independent development and modification of each layer without affecting the overall system. The architecture consists of the following layers:

1. Presentation Layer (Frontend)

The presentation layer is responsible for user interaction and interface rendering. It is developed using **React.js**, which enables component-based development and dynamic rendering of content. This layer handles user inputs such as search queries, chatbot interactions, and navigation between pages.

The frontend communicates with the backend through HTTP requests and displays the data received in a structured format using reusable components like cards, lists, and forms.

2. Application Layer (Backend)

The backend is implemented using **Node.js and Express.js**, which handle all server-side operations. This layer is responsible for processing API requests, executing business logic, and managing communication between the frontend and the database.

The backend defines multiple REST API endpoints such as /api/news, /api/schemes, and /api/search. It processes incoming requests, performs data retrieval using database queries, and returns structured JSON responses to the frontend.

3. Data Layer (Database)

The data layer uses **MongoDB Atlas**, a cloud-based NoSQL database. It stores educational data in multiple collections such as policies, schemes, regulations, and scholarships.

MongoDB's flexible schema design allows efficient storage of unstructured and semi-structured data. The backend interacts with the database using Mongoose models, enabling efficient querying and data manipulation.

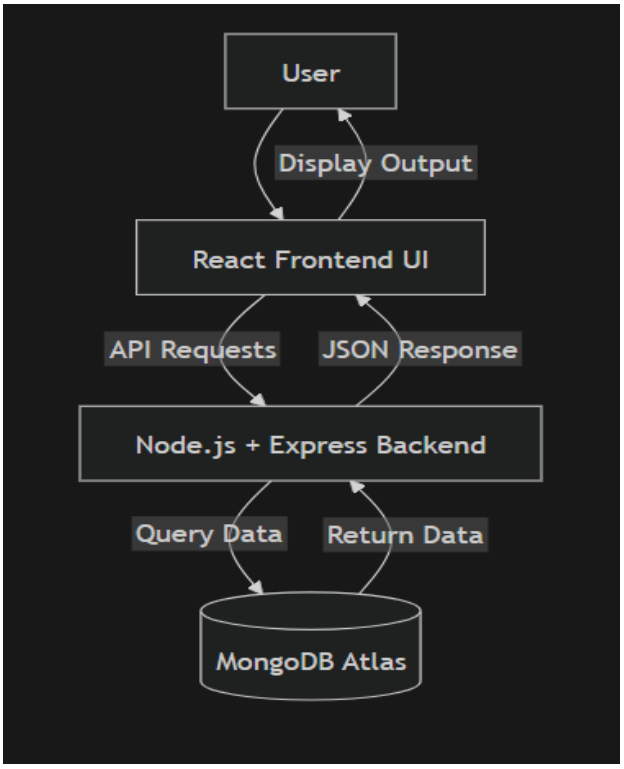


Fig 5.1 System Architecture

5.3 Data Flow Diagram

The Data Flow Diagram (DFD) is used to represent how data moves within the Edu Info Portal system and how different components interact with each other. It provides a clear understanding of the flow of information between the user, frontend, backend, and database. At a high level, the system begins with user interaction through the frontend interface. Users perform actions such as logging in, searching for information, or interacting with the chatbot. These actions generate requests that are sent to the backend server through API calls.

The backend processes these requests by identifying the appropriate route and executing the required logic. For example, when a user performs a search operation, the backend processes the query and retrieves relevant data from multiple collections in the database. The backend then formats the retrieved data into a structured JSON response and sends it back to the frontend.

The frontend receives this data and dynamically renders it using UI components such as cards and lists. This ensures that the user receives real-time and relevant information without reloading the page.

The DFD can be further divided into different levels:

Level 0 (Context Diagram): Represents the entire system as a single process interacting with external entities such as users and the database.

Level 1: Breaks the system into smaller processes such as login, search, chatbot, and data retrieval.

Overall, the data flow mechanism ensures efficient communication between system components.

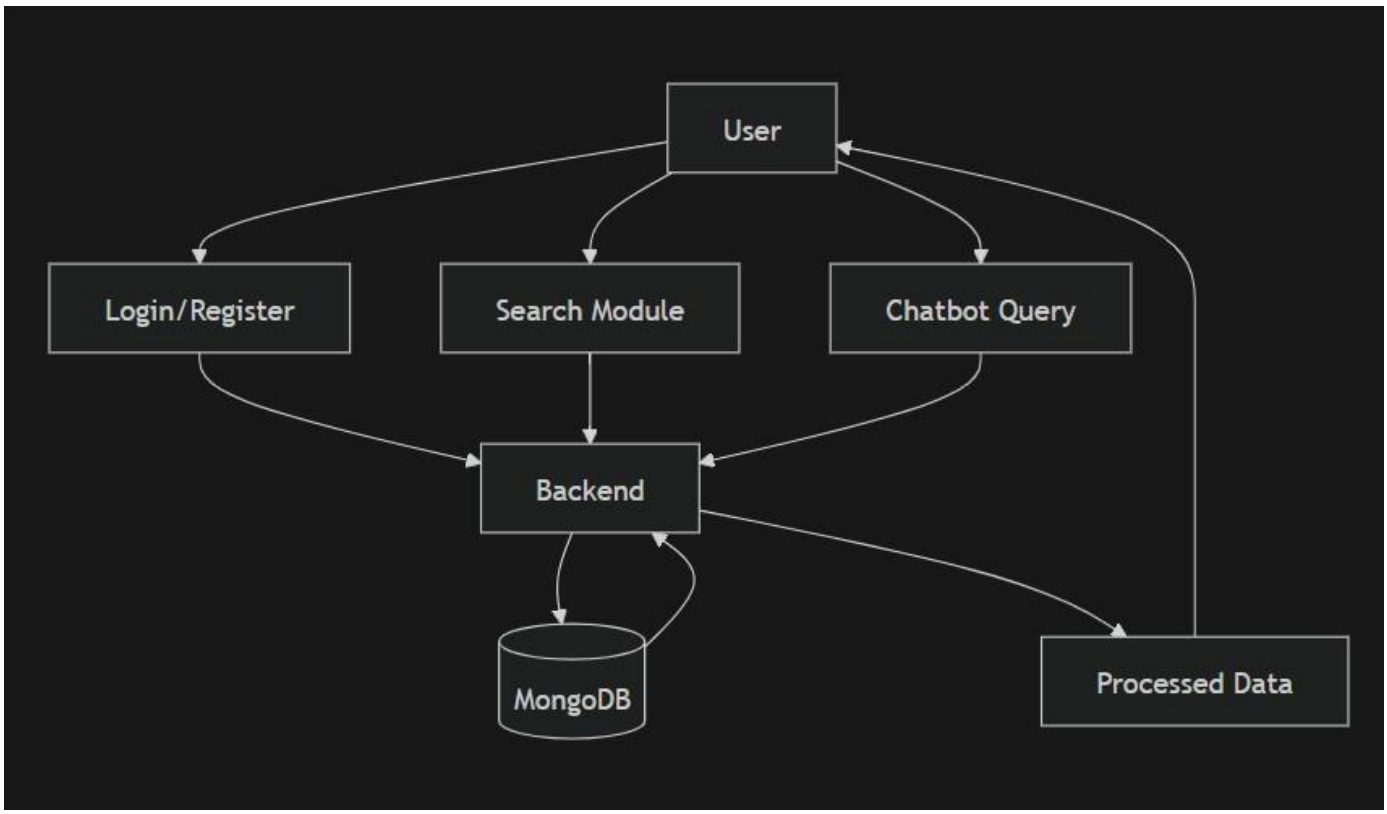


Fig 5.2 Data Flow Diagram

5.4 System Flow Chart

The system flowchart represents the step-by-step operational logic of the Disaster Preparedness and Response Education System, illustrating how user interactions are processed and how the system responds accordingly. It provides a clear visualization of the sequence of actions performed from the moment a user accesses the system to the final output generation.

The flow begins with the user entering the application through the frontend interface developed using React. The user is then presented with multiple options such as selecting a disaster category, attempting a quiz, or interacting with the chatbot. Based on the user's selection, the system processes the request through the backend implemented using Node.js and Express.js.

If the user selects a disaster category, the system retrieves relevant information from the database and displays structured safety guidelines. If the user chooses the quiz module, the system presents questions,

evaluates responses, and displays results. For chatbot interactions, user queries are sent to the AI module powered by Ollama, which generates context-aware responses and returns them to the user interface. The process continues until the user decides to exit the system. The flowchart ensures that all user actions are handled efficiently and logically, providing a seamless and interactive experience.

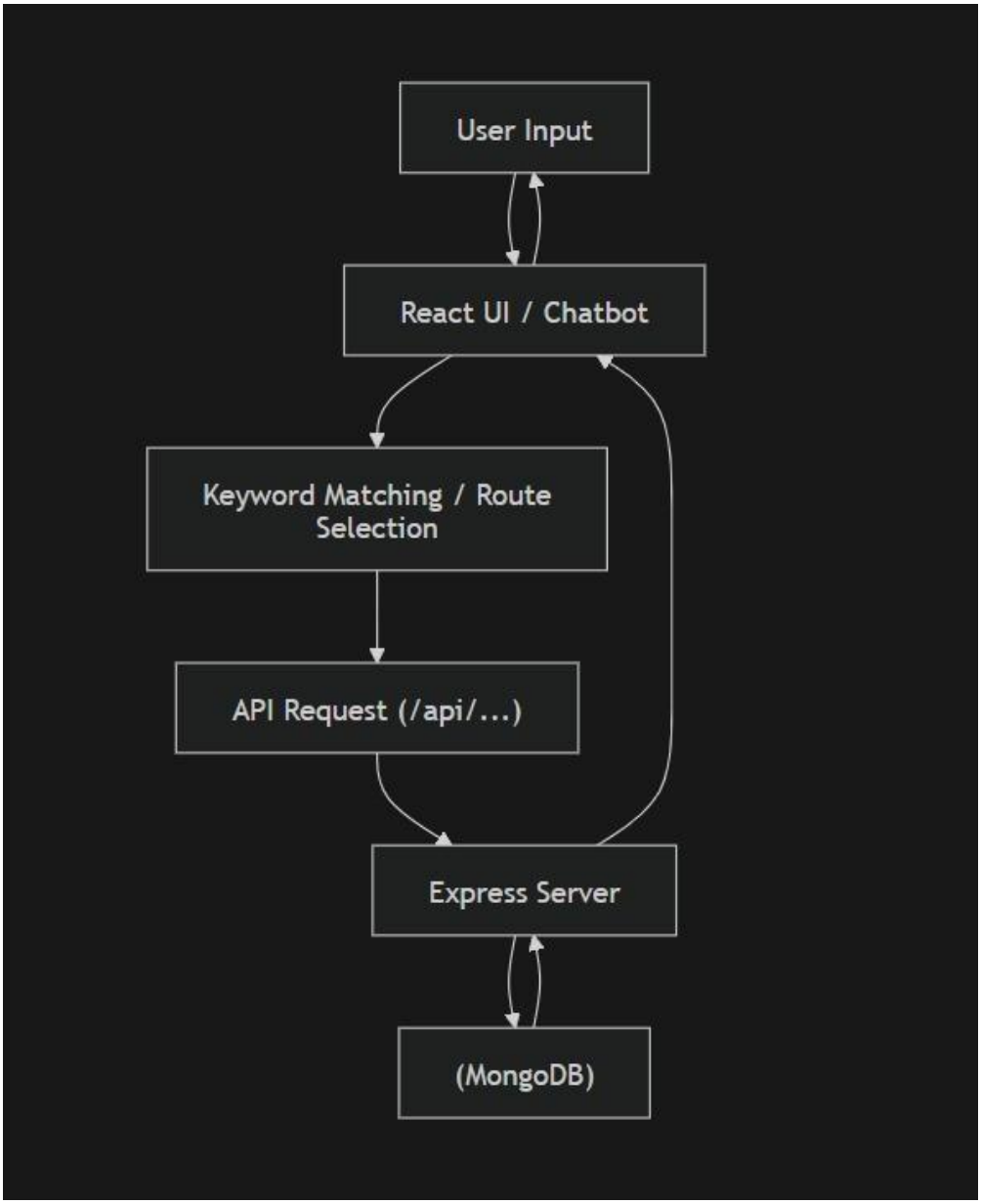


Fig 5.3 Flow Chart

5.5 Module Design

The Edu Info Portal is designed using a modular approach, where the system is divided into smaller functional units called modules. Each module is responsible for performing a specific task, which improves system organization, maintainability, and scalability.

The major modules in the system include:

Authentication Module

This module handles user registration, login, and logout functionalities. It validates user credentials and maintains session state using local storage. It ensures that only authenticated users can access the main features of the application.

Search Module

The search module allows users to find relevant information using keywords. It sends user queries to the backend, where the system performs a search across multiple database collections using pattern matching techniques. This module improves accessibility by reducing the need for manual navigation.

Chatbot Module

The chatbot module acts as an interactive assistant that helps users retrieve information quickly. It uses keyword-based detection to identify user intent and fetch relevant data from the backend. The chatbot enhances user experience by providing a conversational interface.

Content Module

This module is responsible for displaying categorized information such as policies, schemes, and news. It organizes data into structured formats and presents it using UI components like cards and lists.

API Module

The API module manages communication between the frontend and backend. It handles HTTP requests and responses, ensuring efficient data exchange between system components.

Database Module

This module manages data storage and retrieval using MongoDB. It organizes data into collections and ensures efficient querying and scalability.

The modular design ensures that each component operates independently, making it easier to update or enhance specific functionalities without affecting the entire system.

Table 5.1 Functional Requirements

Module	**Description**
Authentication Module	Manages login, registration, and logout
Search Module	Handles keyword-based search
Chatbot Module	Provides quick assistance
Content Module	Displays categorized data
API Module	Handles backend communication
Database Module	Manages data storage

5.6 Database Design

The database design of the Edu Info Portal is implemented using MongoDB Atlas, a NoSQL cloud database. The database is structured into multiple collections, each representing a specific category of educational information.

Unlike traditional relational databases, MongoDB uses a document-based structure, where data is stored in JSON-like format. This allows flexibility in handling different types of data and makes it suitable for dynamic applications.

Each collection contains documents with fields such as title, description, eligibility criteria, issuing authority, and links. This structured format ensures consistency while allowing flexibility for future enhancements.

The use of MongoDB provides several advantages:

- Flexible schema design

- High scalability
- Efficient querying
- Easy integration with Node.js

The database interacts with the backend through Mongoose models, which define the structure and behaviour of data. This ensures data consistency and simplifies database operations.

Table 5.2 Non-Functional Requirements

Collection	**Fields**
Policies	title, description, issuedBy, year, link
Schemes	title, description, eligibility, link
Regulations	title, description, authority
Scholarships	title, description, eligibility
Higher Studies	title, description, country
News	title, description, source, date

5.7 API Design

The Edu Info Portal uses RESTful APIs to enable communication between the frontend and backend. APIs act as intermediaries that process requests and return appropriate responses.

Each API endpoint is designed to handle specific operations such as retrieving data, searching information, or fetching details of a particular item.

For example:

/api/news retrieves educational news

/api/schemes retrieves government schemes

/api/search performs keyword-based search

When a request is made, the backend processes it using Express routes. It then queries the database, retrieves relevant data, and sends it back to the frontend in JSON format.

The API design follows REST principles, ensuring:

Standardized communication

Scalability

Easy integration

Efficient data handling

The search API is particularly important, as it performs multi-collection queries and enables users to access information directly using keywords.

Table 5.3 Model Overview

Method	**Endpoint**	**Description**
GET	/api/news	Fetch news
GET	/api/policies	Fetch policies
GET	/api/schemes	Fetch schemes
GET	/api/regulations	Fetch regulations
GET	/api/higherstudies	Fetch higher studies
GET	/api/search?q=	Perform search

5.8 UI/UX Design

The user interface of the Edu Info Portal is designed with a focus on simplicity, accessibility, and responsiveness. The frontend is developed using React.js and styled using Bootstrap and custom CSS. The design follows a clean layout with intuitive navigation, allowing users to easily access different sections of the portal. Information is displayed using card-based components, which provide a structured and visually appealing presentation.

The search bar is prominently placed to enable quick access to information, while the chatbot provides an interactive way to retrieve data. The login and registration pages are designed with a modern gradient theme to enhance visual appeal.

The system is fully responsive, ensuring compatibility across different devices such as desktops, tablets, and mobile phones.

Overall, the UI/UX design improves user engagement and ensures a seamless browsing experience.

5.9 Technology Stack Justification

The selection of technologies for the Disaster Preparedness and Response Education System is based on factors such as performance, scalability, ease of development, and user experience. The frontend is developed using React, which provides a component-based architecture, enabling efficient development of dynamic and reusable user interface elements. React ensures fast rendering and a smooth user experience, making it suitable for interactive applications.

The backend is implemented using Node.js and Express.js, which offer high performance and efficient handling of multiple requests. Node.js allows asynchronous processing, ensuring faster response times, while Express simplifies API development and server-side logic.

For data storage, MongoDB is used due to its flexibility in handling semi-structured data. It supports scalable and efficient storage of disaster information, quiz data, and chatbot interactions. The document based structure of MongoDB aligns well with the dynamic nature of the application.

A key component of the system is the AI chatbot powered by Ollama. Ollama enables local execution of language models, ensuring enhanced privacy, reduced latency, and independence from external APIs. Overall, this technology stack provides a robust, scalable, and efficient foundation for developing a responsive and intelligent disaster preparedness system.

Table 5.4 Database Collections and Fields

Technology	Reason
React.js	Efficient UI rendering
Node.js	Fast backend processing
Express.js	Simplified API development
MongoDB Atlas	Scalable database
Vercel	Easy frontend deployment
Render	Reliable backend hosting

5.10 Security & Performance

The Disaster Preparedness and Response Education System is designed with a strong focus on security and performance to ensure reliable and safe user interaction. Security measures are implemented to protect the system from unauthorized access and malicious activities. Input validation is performed at both frontend and backend levels to prevent common vulnerabilities such as injection attacks and invalid data submission. Sensitive data handling is minimized, and chatbot interactions powered by Ollama are processed locally, ensuring better privacy and reducing exposure to external threats.

Authentication mechanisms can be implemented if user-specific features are added in the future, ensuring that only authorized users access certain functionalities. Proper error handling and logging are also incorporated to track system activities and identify potential issues.

Performance optimization is equally important to provide a smooth and responsive user experience. The system is designed to deliver fast response times by using efficient API handling through Node.js and Express.js. Database queries in MongoDB are optimized to ensure quick data retrieval. The use of asynchronous processing in Node.js helps handle multiple user requests simultaneously without delays.

5.11 Deployment Architecture

The deployment architecture of the Disaster Preparedness and Response Education System defines how the application components are hosted and executed in a real-world environment. The system follows a

client-server architecture where the frontend, backend, database, and AI chatbot are deployed either locally or on cloud platforms to ensure accessibility and performance.

The frontend application developed using React is deployed on a web hosting platform or served locally using development servers. It acts as the user-facing layer, allowing users to interact with the system through a browser. The backend, implemented using Node.js and Express.js, is hosted on a server that handles API requests, business logic, and communication with the database.

The database, managed using MongoDB, can be deployed locally or on cloud services to store disaster information, quiz data, and system content. A key aspect of the deployment is the AI chatbot powered by Ollama, which runs locally on the server or a dedicated machine to ensure privacy and low latency. This architecture allows flexibility in deployment, supporting both local development and scalable cloudbased hosting. It ensures efficient communication between components and provides a reliable and responsive user experience.

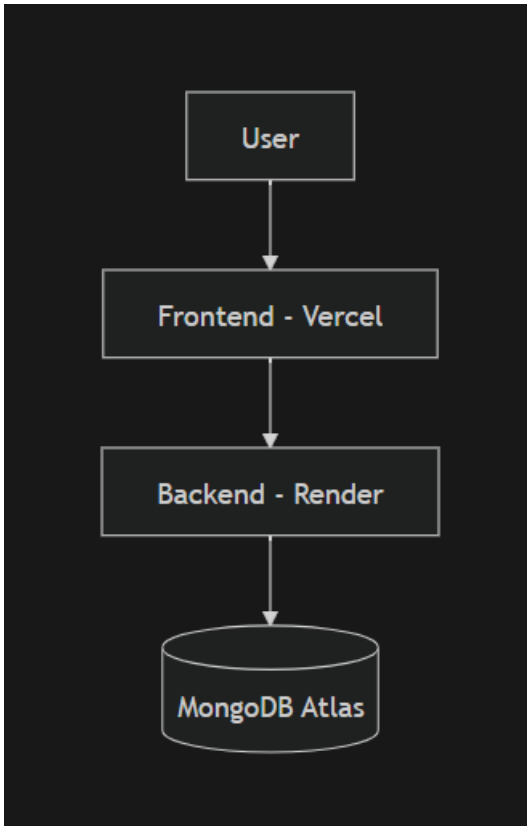


Fig 5.4 Architecture of Deployment

5.12 Constraints & Assumptions

The design and implementation of the Disaster Preparedness and Response Education System are subject to certain constraints and assumptions that influence its functionality and scope. These factors are important to consider for understanding the limitations of the system and planning future improvements. One of the primary constraints is the dependency on internet connectivity for accessing the web-based frontend and backend services. Although the AI chatbot powered by Ollama can operate locally, users still require a stable connection to access the application interface and retrieve data. Another limitation is the restricted coverage of disaster types, as the system currently focuses on selected categories such as earthquakes, floods, and fires. Expanding to more disaster scenarios may require additional data collection and system updates.

The system also assumes that users have basic digital literacy and can navigate a web-based platform. Device compatibility is considered, but performance may vary depending on hardware capabilities, especially when running local AI models. Additionally, the accuracy of information depends on the quality of data collected from external sources, and periodic updates are necessary to maintain reliability. From a development perspective, the system assumes the availability of necessary tools and technologies such as React, Node.js, and MongoDB.

The system operates under certain limitations:

- It uses free-tier cloud services, which may have performance limits
- Authentication is basic and not suitable for high-security applications
- The system does not support real-time updates
- Advanced AI-based features are not implemented

The system is developed based on the following assumptions:

- Users have access to the internet
- The data stored in the database is accurate and updated
- The system is primarily used for educational purposes
- Users have basic familiarity with web applications

Chapter 6

Hardware, Software and Simulation

This chapter describes the hardware and software requirements used for the development and execution of the Edu Info Portal. It also explains the implementation and testing environment used to simulate the system functionality.

The project is developed as a web-based application, which minimizes the need for specialized hardware and allows it to run on standard computing systems.

6.1 Hardware

Table 6.1 Hardware Requirements

Component	**Specification**
Processor	Intel Core i3 or above
RAM	Minimum 4 GB (8 GB recommended)
Storage	At least 10 GB free space
System Type	Laptop/Desktop
Internet	Stable internet connection

6.2 Software Development Tools

The development of the Edu Info Portal involves modern web technologies and tools that support efficient development, testing, and deployment.

Table 6.2 Software Development Tools

Category	Tool / Technology	Purpose
Code Editor	Visual Studio Code	Writing and managing source code
Frontend	React.js + Bootstrap	Building responsive UI
Backend	Node.js + Express.js	Server-side logic and APIs
Database	MongoDB Atlas	Cloud-based data storage
Version Control	Git & GitHub	Code management and collaboration
Deployment (Frontend)	Vercel	Hosting React application
Deployment (Backend)	Render	Hosting backend server
Browser	Google Chrome	Testing and debugging
Package Manager	npm	Managing dependencies

6.3 Implementation

The Edu Info Portal is implemented using a full-stack web development approach, where frontend and backend are developed independently and integrated through APIs.

Frontend Implementation

- Developed using React.js
- Uses reusable components (Navbar, Pages, Chatbot)
- Implements routing using React Router
- Styled using Bootstrap and custom CSS
- Handles user interaction and navigation

Backend Implementation

- Developed using Node.js and Express.js
- Provides REST API endpoints
- Handles data retrieval and processing
- Connects to MongoDB Atlas database
- Manages routes for schemes, policies, news, etc.

Authentication Implementation

- Login and Register functionality implemented
- User data stored in LocalStorage (current stage)
- Session handled using isLoggedIn flag
- Logout clears session and redirects to login

The system follows a modular implementation approach, where each component is developed independently and then integrated. This improves maintainability and scalability.

The frontend communicates with the backend using API calls, ensuring dynamic data loading. Authentication is currently handled using local storage, which can later be enhanced using JWT and database-based authentication.

6.4 Simulation

Simulation is the process of testing the system in controlled environments to ensure correct functionality before deployment.

Simulation Process

1. Development and testing on local environment (localhost)
2. Testing UI responsiveness across devices
3. Verifying navigation between pages
4. Testing login and registration functionality
5. Checking chatbot interaction
6. Testing API responses from backend
7. Verifying database connectivity (MongoDB Atlas)
8. Deployment on Vercel (frontend) and Render (backend)
9. Testing live application through deployed URL

Chapter 7

Evaluation and Results

This chapter presents the evaluation of the Edu Info Portal system through structured testing and analysis. The purpose of this phase is to verify whether the developed system meets the functional and non-functional requirements defined earlier, and to assess its overall performance, usability, and reliability. The evaluation is carried out by defining test points, preparing a test plan, executing test cases, and analyzing the results obtained from system usage.

7.1 Test points

Test points represent the key functionalities of the system that were evaluated during testing. These points are derived from the core features of the Edu Info Portal and ensure that all critical components function as expected.

To systematically evaluate the system, multiple test scenarios were defined covering authentication, navigation, search, chatbot interaction, and API functionality.

Table 7.1 Test Points

Test Point ID	Feature	Description
TP1	User Authentication	Validate login and registration functionality
TP2	Navigation	Ensure smooth navigation across pages
TP3	Search Functionality	Verify search results based on user query
TP4	Data Display	Check correct rendering of content
TP5	Chatbot	Validate chatbot responses
TP6	Logout Functionality	Ensure proper session termination

TP7	API Integration	Verify frontend-backend communication
TP8	Database Connectivity	Ensure data retrieval from MongoDB

7.2 Test Plan

The test plan outlines the approach used to evaluate the system. It includes the type of testing performed, test environment, and execution strategy.

Testing Types Used

- Functional Testing
- Integration Testing
- UI/UX Testing
- System Testing

Table 7.2 Test Plan

Component	**Environment**
Frontend	React (Vite) on browser
Backend	Node.js server (Render)
Database	MongoDB Atlas
Browser	Google Chrome
OS	Windows

Test Execution Strategy

1. Test each module independently (Unit Testing)
2. Integrate frontend and backend and test APIs
3. Perform user-level testing (end-to-end flow)
4. Test deployment version (live environment)
5. Fix bugs and re-test

7.3 Test Results

The test results represent the outcome of executing the defined test cases under various conditions. The Edu Info Portal was tested systematically to ensure that all functional components operate correctly and provide the expected outputs.

Testing was conducted in both the **local development environment** and the **deployed cloud environment** (Vercel and Render). This ensured that the system behaves consistently in real-world usage conditions. The results indicate that the system performs reliably across all major modules, including authentication, navigation, search, chatbot, and API integration.

Functional Testing Results

The authentication system was tested using both valid and invalid credentials. When correct credentials were provided, the system successfully redirected the user to the home page. In cases of incorrect input, appropriate error messages were displayed, ensuring proper validation.

The navigation functionality was tested by accessing different sections of the portal such as policies, schemes, and higher studies. All routes were correctly handled, and pages were loaded without errors, indicating proper frontend routing implementation.

The search functionality was one of the key features evaluated. When users entered keywords such as "UGC" or specific scheme names, the system successfully fetched relevant results from the backend. The results were displayed dynamically, demonstrating correct API integration and database querying.

The chatbot module was tested with various user queries. The system responded appropriately by identifying keywords and guiding users to relevant sections. This confirmed that the chatbot logic is functioning as intended.

Integration Testing Results

Integration testing was performed to verify communication between frontend, backend, and database components. API endpoints were tested to ensure that data is correctly retrieved from MongoDB Atlas and displayed on the frontend.

All API calls returned valid responses, and no major data inconsistencies were observed. This confirms that the system integration is stable and efficient.

System Testing Results

The complete system was tested as a whole to evaluate end-to-end functionality. Users were able to:

- Register and login
- Navigate through different modules
- Search for information
- Interact with the chatbot
- Logout successfully

The system handled all operations smoothly, with minimal response delay.

Table 7.3 Test Result Summary

Feature	Result
Authentication	Working correctly
Navigation	Smooth and error-free
Search	Accurate results displayed
Chatbot	Responding correctly
API Integration	Stable and reliable
Database Connectivity	Successful data retrieval

7.4 Insights

The evaluation of the system provided several important insights regarding its performance, usability, and overall effectiveness.

Key Observations

One of the major observations is that the centralized structure of the portal significantly reduces the effort required by users to access educational information. Instead of visiting multiple websites, users can retrieve all relevant data from a single platform.

The search functionality proved to be highly effective in improving user experience. Users can directly access specific information without navigating through multiple pages, which enhances efficiency.

The chatbot feature adds an interactive layer to the system. It assists users in quickly finding relevant information, especially for users who are unfamiliar with the platform.

The modular architecture of the system ensures smooth interaction between components. The frontend, backend, and database communicate efficiently, resulting in fast data retrieval and minimal delays.

User Experience Insights

The system provides a clean and intuitive user interface, making it easy for users to navigate. The use of card-based layouts and categorized sections improves readability and organization.

Users can easily understand how to use the system without requiring technical knowledge, which makes the platform accessible to a wider audience.

Performance Insights

The system performs efficiently under normal usage conditions. API response times are quick, and data is displayed dynamically without requiring page reloads.

The use of cloud-based services ensures that the system remains accessible and scalable.

Limitations Identified

Despite the successful implementation, certain limitations were observed:

- Authentication is currently handled using **local storage**, which is not fully secure
- Chatbot responses are **keyword-based** and not intelligent
- Search functionality currently shows limited results (can be expanded)
- No role-based access control (admin/user separation not implemented)

Recommendations for Improvement

Based on the insights obtained, the following improvements are suggested:

- Implement **JWT-based authentication** with backend validation
- Upgrade chatbot using **AI/ML techniques**
- Enhance search with **multiple results and filtering options**
- Add **admin dashboard** for managing data
- Improve security using encryption and secure session handling

Chapter 8

Social, Legal, Ethical, Sustainability and Safety Aspects

The development and deployment of the Edu Info Portal involve several important considerations beyond technical implementation. These include social impact, legal compliance, ethical responsibilities, sustainability, and user safety. This chapter discusses how the system aligns with these aspects and highlights its broader implications.

8.1 Social Aspects

The Edu Info Portal contributes positively to society by improving access to educational information. In many cases, students face difficulties in finding reliable and structured information about scholarships, policies, and higher education opportunities. The system addresses this issue by providing a centralized platform where users can access all relevant information in one place.

By reducing dependency on multiple sources, the portal saves time and improves awareness among students. It is especially beneficial for users from rural or underprivileged backgrounds who may not have easy access to multiple educational resources.

The platform promotes digital inclusion by making educational information easily accessible through a web-based interface. This helps bridge the information gap and supports equal opportunities for all users.

8.2 Legal Aspects

The Edu Info Portal operates within the scope of publicly available educational information. The data displayed on the platform is sourced from official or publicly accessible resources, ensuring that no confidential or restricted information is used.

The system adheres to basic legal considerations such as:

- Avoiding unauthorized data usage
- Providing proper references or links where applicable
- Ensuring that content does not violate intellectual property rights

Since the current system does not store sensitive personal data beyond basic login information, the risk of legal issues related to data privacy is minimal. However, future implementations involving database-based authentication must comply with data protection regulations.

8.3 Ethical Aspects

Ethical considerations play an important role in ensuring that the system is used responsibly. The Edu Info Portal is designed to provide accurate and unbiased information to users.

The system ensures:

- Transparency in information presentation
- No misleading or manipulated data
- Equal access to all users without discrimination

The chatbot and search functionalities are designed to assist users without influencing their decisions. The platform aims to empower users by providing reliable information rather than promoting specific outcomes.

8.4 Sustainability Aspects

The Edu Info Portal is developed using modern web technologies and cloud-based services, which contribute to sustainability in multiple ways.

Since the system is digital, it reduces the need for physical resources such as paper and printed materials. This contributes to environmental sustainability by minimizing waste.

The use of cloud platforms such as MongoDB Atlas, Vercel, and Render ensures efficient resource utilization and scalability. These platforms optimize energy usage compared to traditional on-premise systems.

The modular architecture of the system allows easy updates and expansion, ensuring long-term usability without requiring complete redevelopment.

8.5 Safety Aspects

Safety in web applications primarily refers to data security and safe user interaction. The Edu Info Portal implements basic safety measures to ensure a secure user experience.

These include:

- Input validation to prevent invalid data entry
- Controlled navigation to avoid broken or unsafe links
- Proper error handling to prevent system crashes

Currently, user authentication is managed using local storage, which provides basic session handling. However, for enhanced safety, future improvements may include secure authentication mechanisms such as JWT and encryption.

The system does not expose sensitive user data, reducing the risk of data breaches.

Chapter 9

Conclusion

9.1 Conclusion

The Edu Info Portal has been successfully developed as a centralized web-based platform aimed at improving access to educational information. The system addresses the major challenge of fragmented data sources by integrating multiple categories such as policies, schemes, regulations, scholarships, higher studies, and educational news into a single unified interface.

The project demonstrates the effective use of modern full-stack web technologies, including React.js for building a dynamic and responsive user interface, Node.js and Express.js for backend processing, and MongoDB Atlas for cloud-based data storage. This combination enables efficient data handling, scalability, and seamless communication between system components.

The platform provides several user-oriented features such as login and registration, smart search functionality, and a chatbot interface. These features enhance user interaction and significantly reduce the effort required to locate relevant information. The system is also deployed using cloud platforms such as Vercel and Render, ensuring accessibility from any location.

The evaluation results confirm that the system performs reliably across all modules and meets the intended objectives. Users can easily navigate the platform, retrieve information, and interact with the system without complexity. Overall, the Edu Info Portal serves as an effective solution for improving accessibility, organization, and usability of educational information.

9.2 Key Contributions

The Edu Info Portal makes several important contributions in both technical and practical aspects. One of the primary contributions is the development of a **centralized information system** that consolidates multiple categories of educational data into a single platform. This reduces the dependency on multiple websites and simplifies the process of accessing information.

The project also demonstrates the practical implementation of a full-stack MERN architecture, showcasing how frontend, backend, and database components can be effectively integrated. This highlights the scalability and modularity of modern web application development.

Another key contribution is the implementation of smart search functionality, which allows users to directly access relevant information using keywords. This improves efficiency and enhances the overall user experience.

The inclusion of a chatbot interface adds an interactive dimension to the system, enabling users to quickly retrieve information without navigating through multiple pages.
Additionally, the project demonstrates the use of cloud-based deployment, making the system accessible globally. This reflects real-world application development practices and enhances the practicality of the system.

9.3 Limitations of the System

Despite its successful implementation, the current version of the Edu Info Portal has certain limitations. The authentication system is implemented using **local storage**, which provides only basic session management and is not suitable for secure applications. A more robust authentication mechanism is required for production-level systems.

The chatbot functionality is based on **keyword matching**, which limits its ability to understand complex queries. It does not use natural language processing or machine learning techniques, which could improve its intelligence and responsiveness.

The search functionality currently returns limited results and does not support advanced filtering or ranking mechanisms. This may affect usability when the dataset becomes larger.

The system also lacks features such as **real-time notifications**, **user personalization**, and **role-based access control**, which are important for enhancing user engagement and system security.

Additionally, the database currently contains a limited set of data, and the system relies on manual data entry, which can restrict scalability.

9.4 Future Work

The Edu Info Portal provides a strong foundation for future enhancements and can be extended in multiple ways to improve functionality and performance.

One of the major improvements would be the implementation of **secure authentication mechanisms** such as JWT (JSON Web Tokens) and database-based user management. This would enhance security and allow better session handling.

The chatbot can be upgraded to an **AI-based intelligent assistant** using natural language processing (NLP) techniques. This would enable the system to understand complex user queries and provide more accurate responses.

The search functionality can be enhanced by adding **advanced filtering options**, **multiple result displays**, and **ranking algorithms** to improve relevance and usability.

An **admin dashboard** can be developed to allow authorized users to add, update, or delete data directly from the system, eliminating the need for manual database updates.

The system can also include **real-time notifications** to inform users about new policies, scholarships, or updates. Additionally, personalization features can be added to recommend content based on user preferences.

Further improvements may include mobile application support, integration with external APIs, and optimization for handling large-scale data.

References

1) R. Kitchin, "The real-time city? Big data and smart urbanism," *GeoJournal*, 2014.
2) Anthopoulos, "Digital governance and smart information systems," *IEEE Access*, 2022
3) T. Erl, *Service-Oriented Architecture: Concepts, Technology, and Design*, Pearson, 2005.
4) M. Fowler, *Patterns of Enterprise Application Architecture*, Addison-Wesley, 2003.
5) J. W. Satzinger, R. B. Jackson, and S. D. Burd, *Systems Analysis and Design in a Changing World*, 7th ed., Cengage, 2019.
6) S. Alter, *Information Systems: Foundation of E-Business*, Prentice Hall, 2002.
7) Amazon Web Services, "Scalable Web Application Architecture," AWS Whitepaper, 2022.
8) Google Cloud, "Web Application Architecture Best Practices," 2023.
9) Banks and E. Porcello, *Learning React*, O'Reilly Media, 2020.
10) Cantelon, M. Harter, T. Holowaychuk, and N. Rajlich, *Node.js in Action*, Manning, 2017.
11) Alammary, "Blended learning models for educational institutions," *TechTrends*, 2019.
12) M. Ally and B. Khan, *International Handbook of E-Learning*, Routledge, 2020.
13) P. D. Gupta and R. Bansal, "Web-based educational information systems for student services," *IJCA*, 2019.
14) S. Kumar and R. Singh, "Centralized educational portals for improving access," *IJIS*, 2021.
15) Y. Chen, L. Zhang, and H. Wang, "Scalable web architectures for information retrieval systems," *IEEE Internet Computing*, 2020.

Base Paper

The development of the Edu Info Portal is based on concepts derived from research in web-based information systems, centralized data platforms, and scalable web architectures.

Gupta and Bansal [13] discuss web-based educational information systems that improve student services by providing centralized access to academic data. Their work highlights the importance of structured information delivery, which forms the foundation of this project.

Kumar and Singh [14] emphasize the role of centralized educational portals in reducing information fragmentation and improving accessibility. Their research supports the need for a unified system that aggregates data from multiple sources into a single platform.

Chen et al. [15] present scalable web architectures for information retrieval systems, focusing on efficient data access and performance optimization. These concepts are reflected in the Edu Info Portal through the use of RESTful APIs and modular system design.

Additionally, studies on modern web technologies such as React [9] and Node.js [10] provide the technical foundation for building dynamic, scalable, and interactive web applications. Cloud-based deployment models discussed in [7] and [8] further support the implementation of the system using platforms like MongoDB Atlas, Render, and Vercel.

Thus, the Edu Info Portal is developed by integrating principles from centralized information systems, scalable architectures, and modern web development technologies to create an efficient and user-friendly educational platform.

Appendix - A Project Repository

GitHub Link: https://github.com/AadhishPranav/digital-information-management-portal

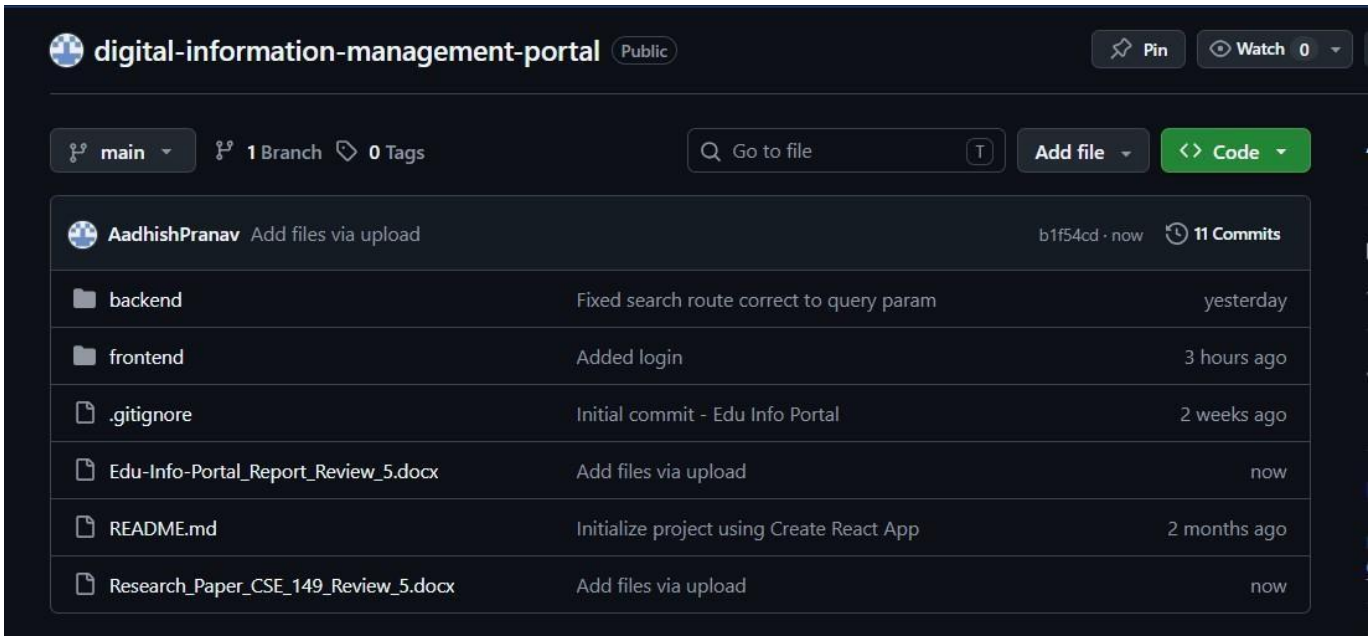


Fig A. GitHub Repository

Appendix – B

Few Images of the Project

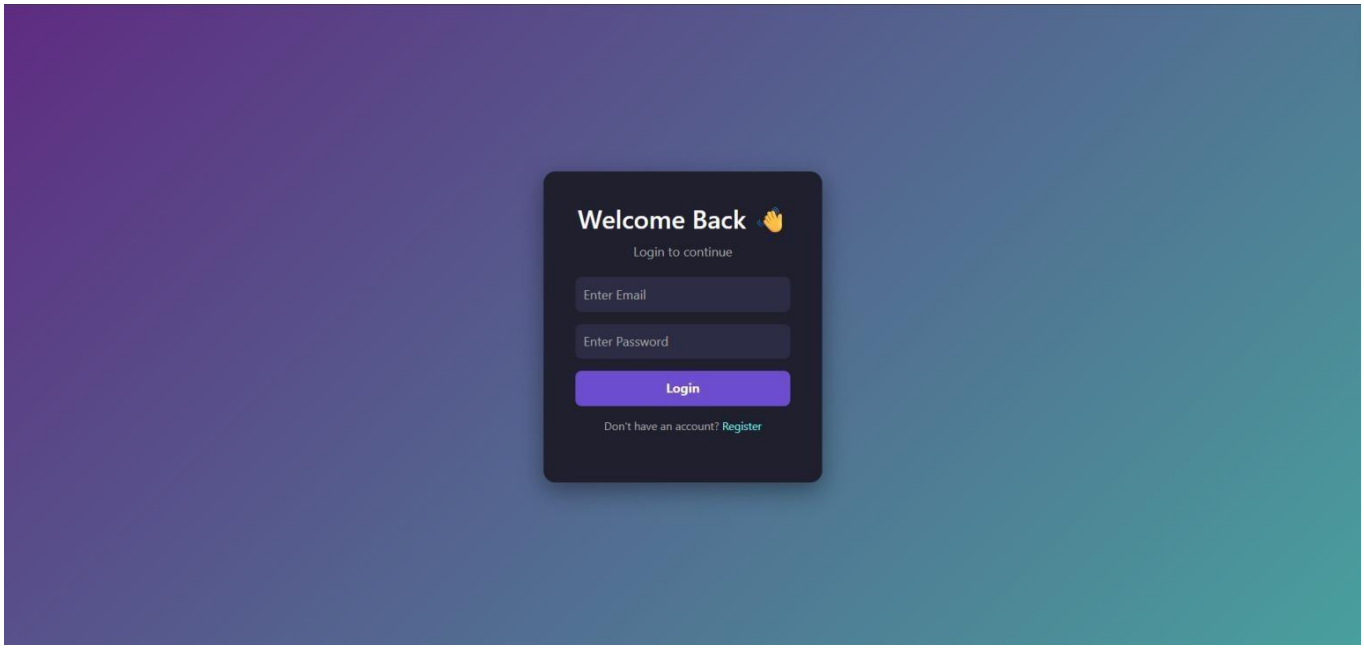


Fig B. Website Login Page

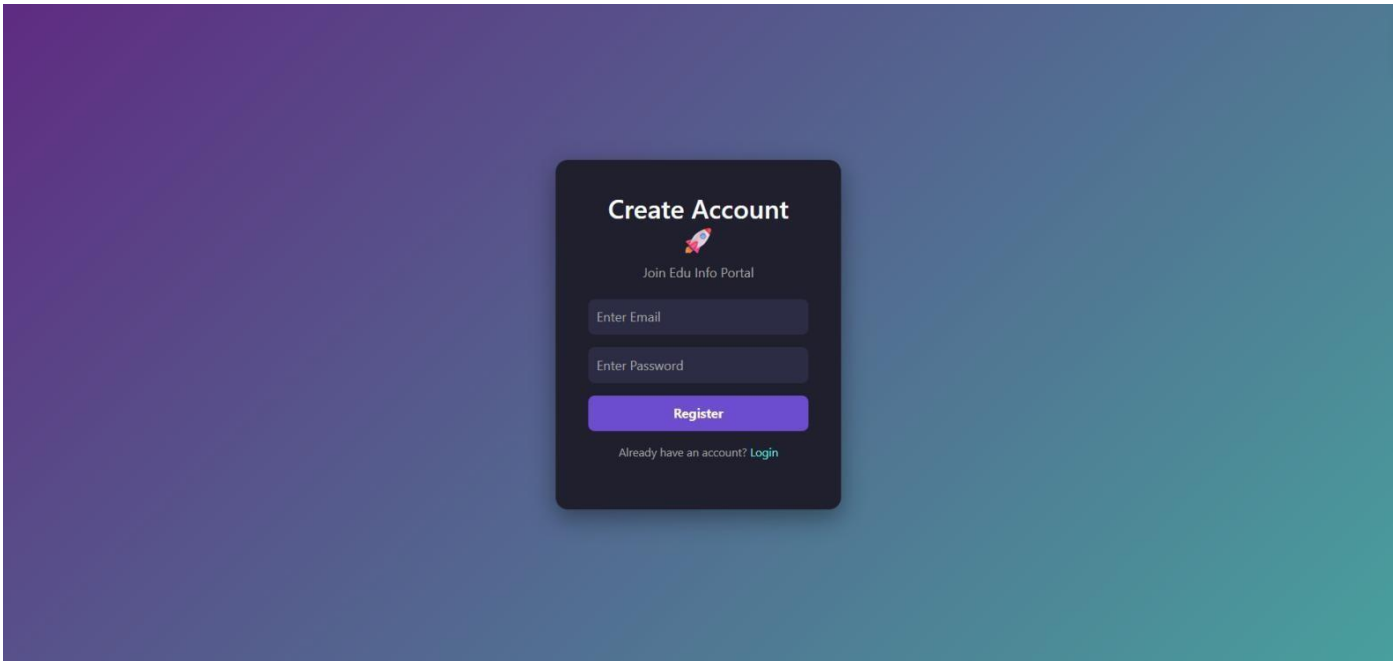


Fig C. Register Page

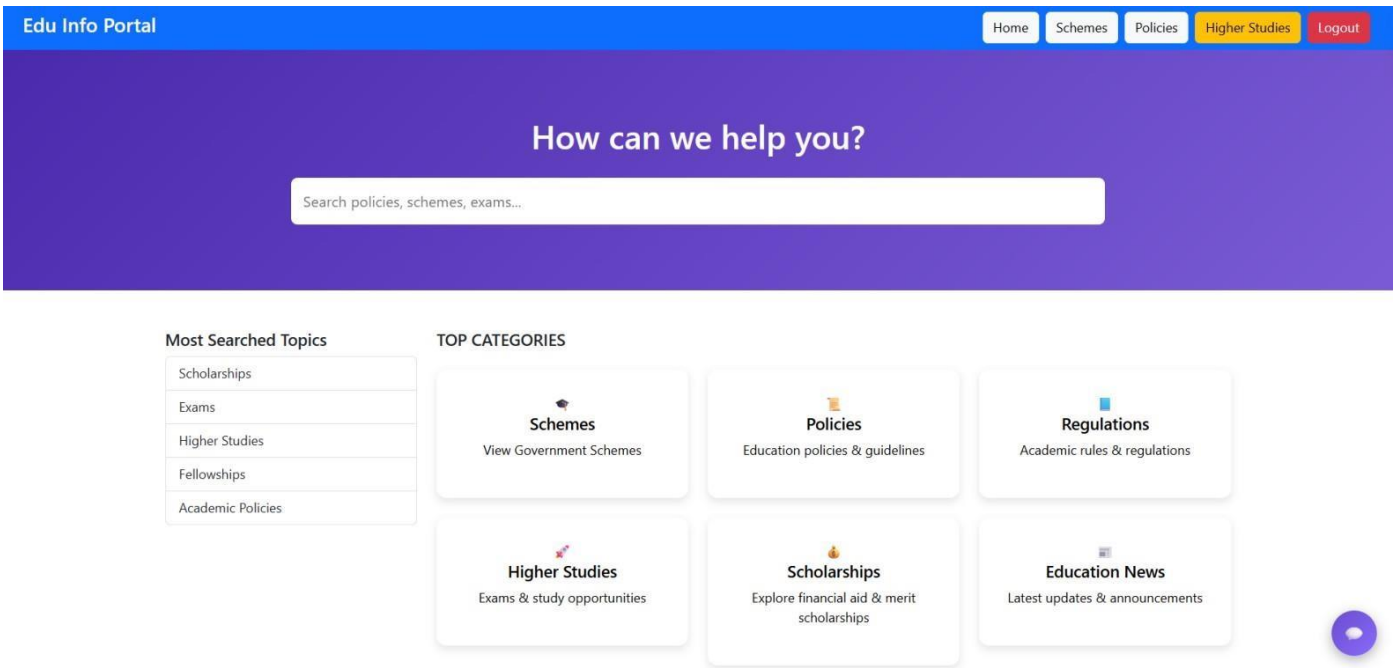


Fig D. Home Page

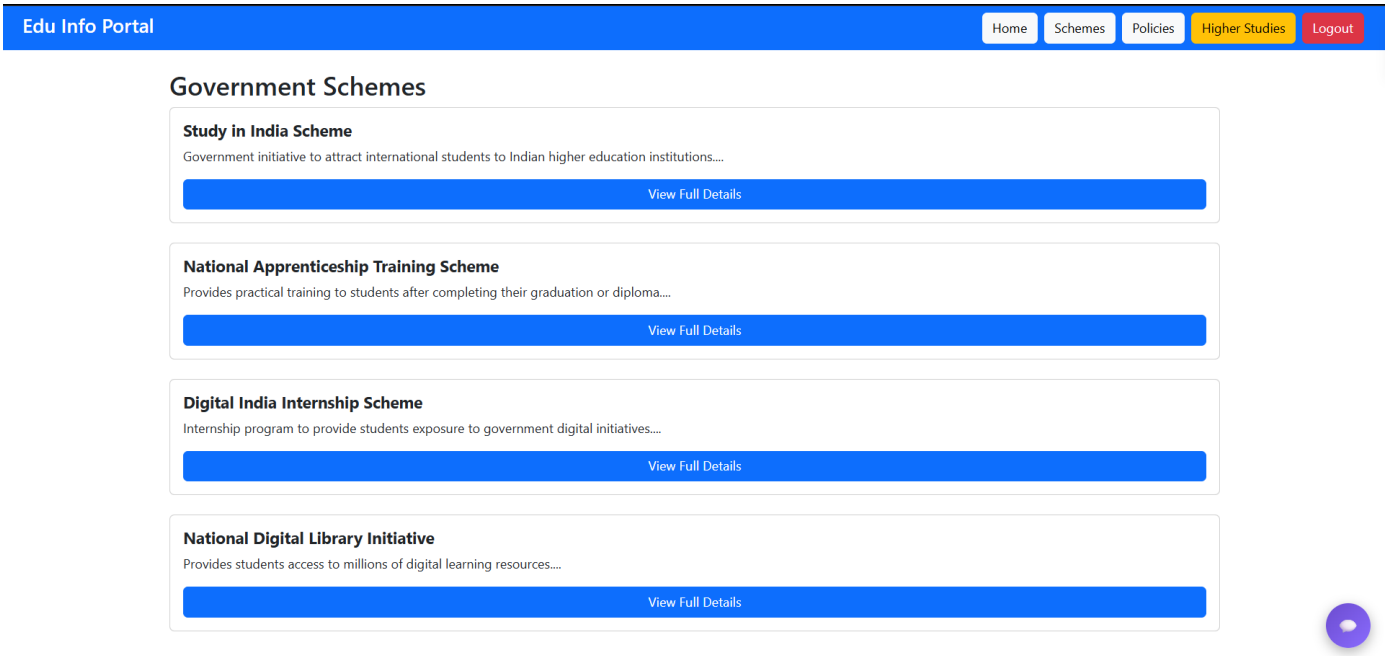


Fig E. Schemes Dashboard

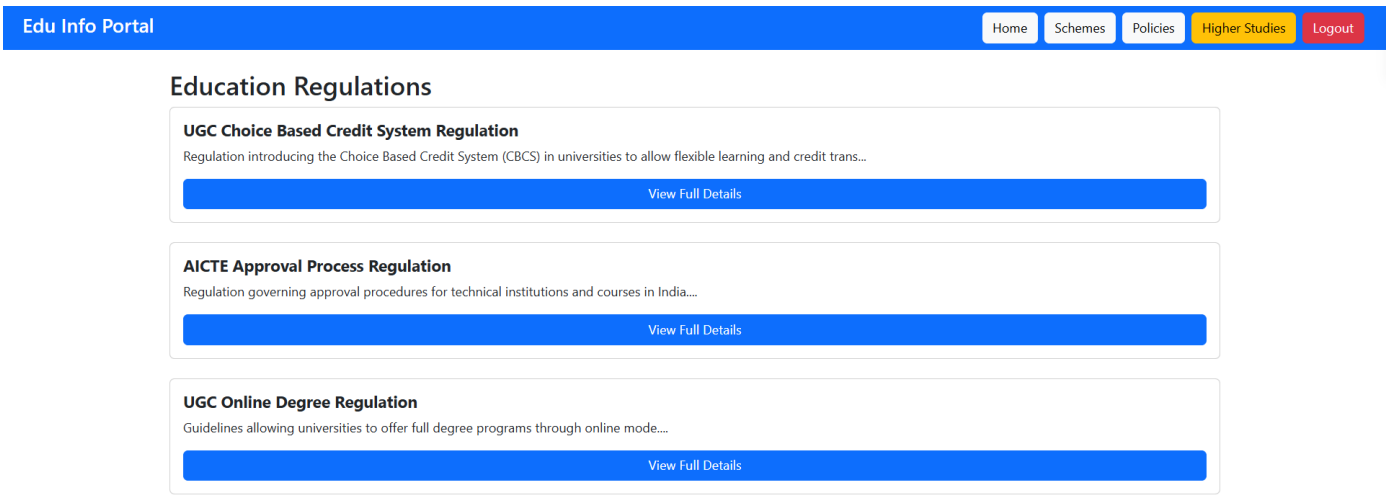


Fig F. Regulations Dashboard

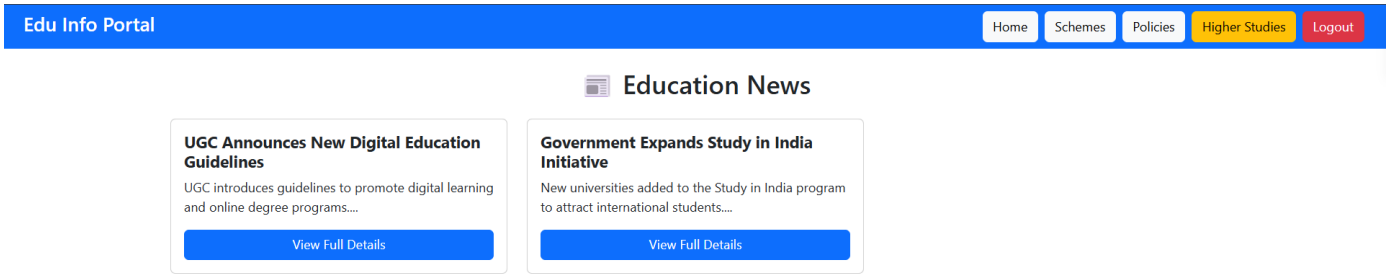


Fig G. News Dashboard

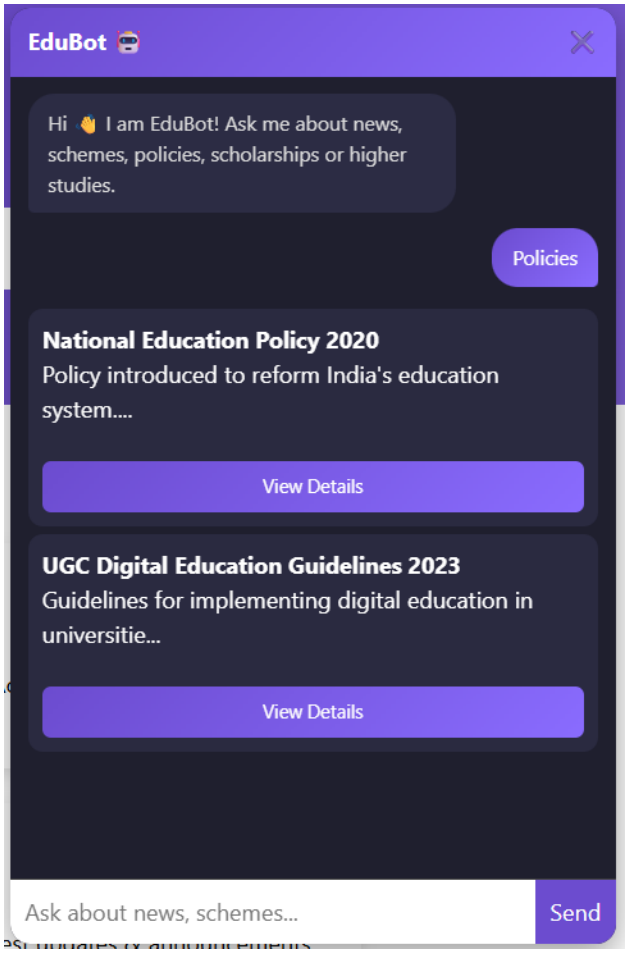


Fig H. Chatbot Response

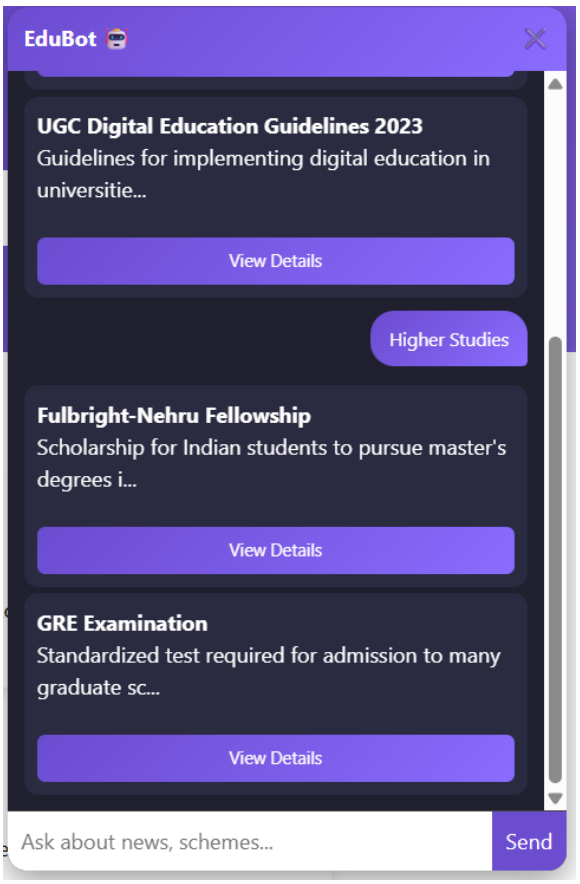


Fig I. Chatbot Response

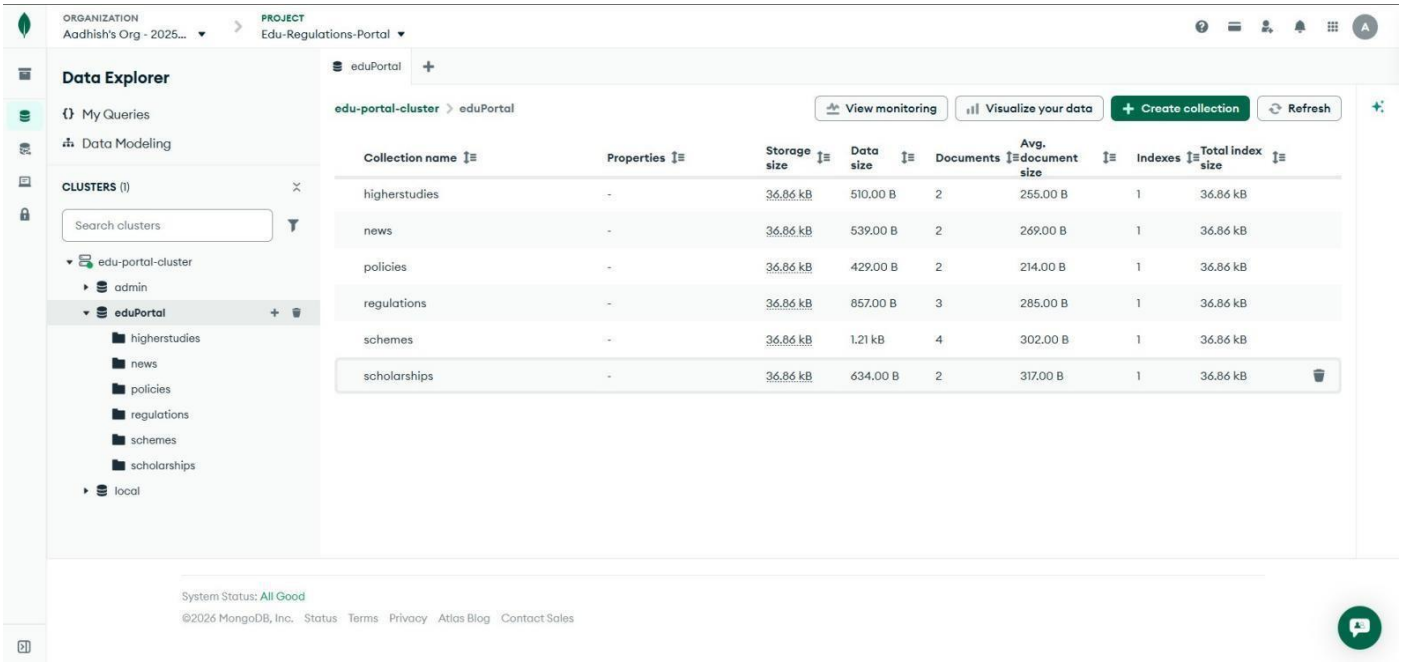


Fig J. Database Mongo DB Atlas

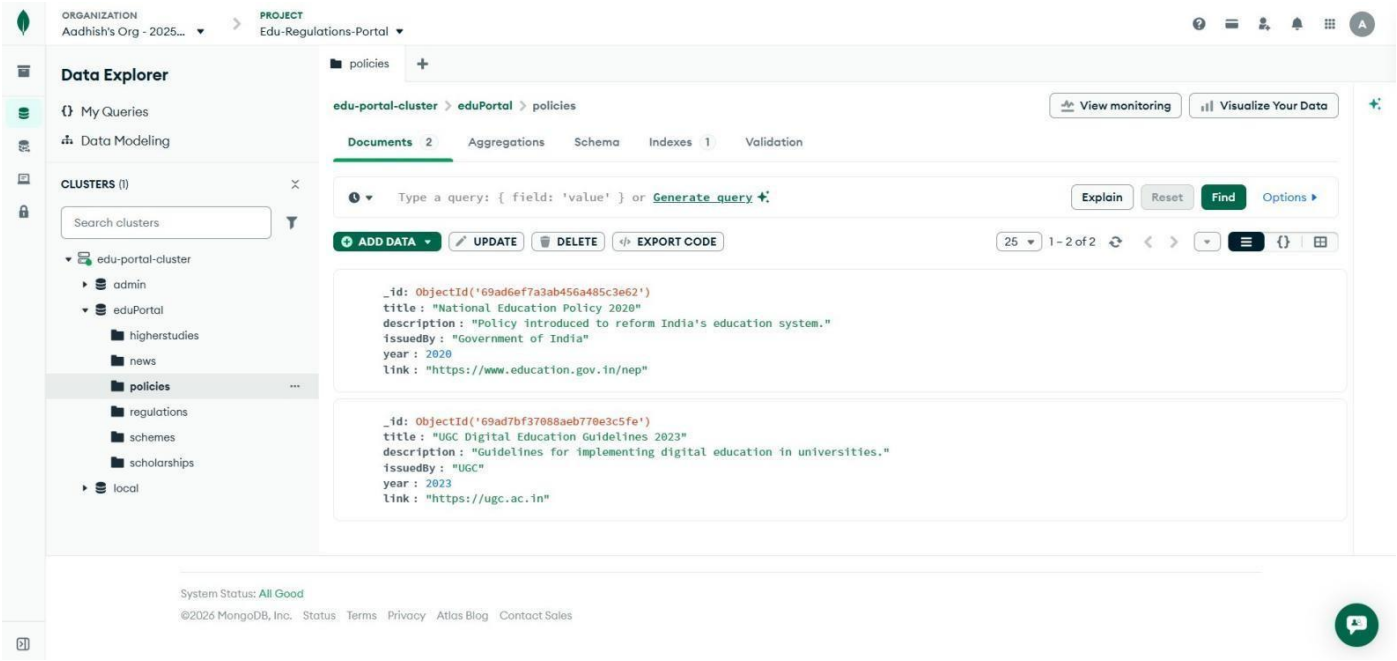


Fig K. MongoDB Atlas – Policies Collection Data

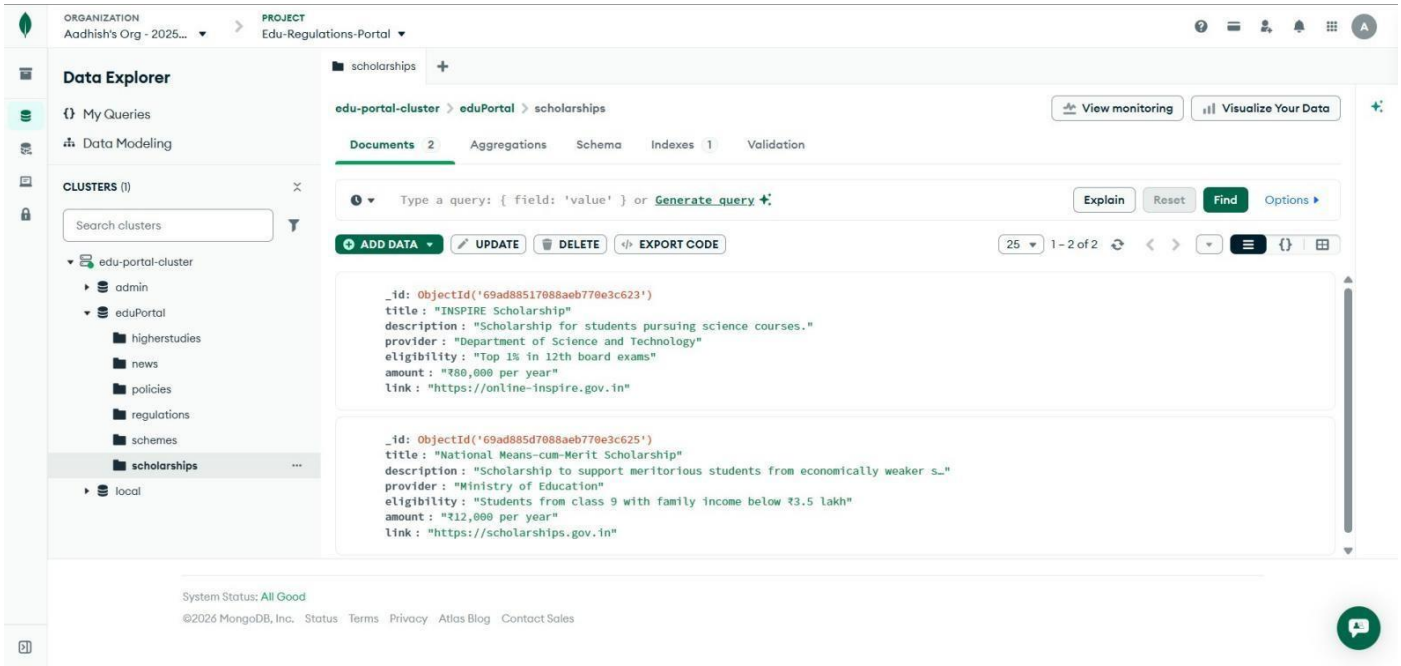


Fig L. MongoDB Atlas – Scholarships Collection Data

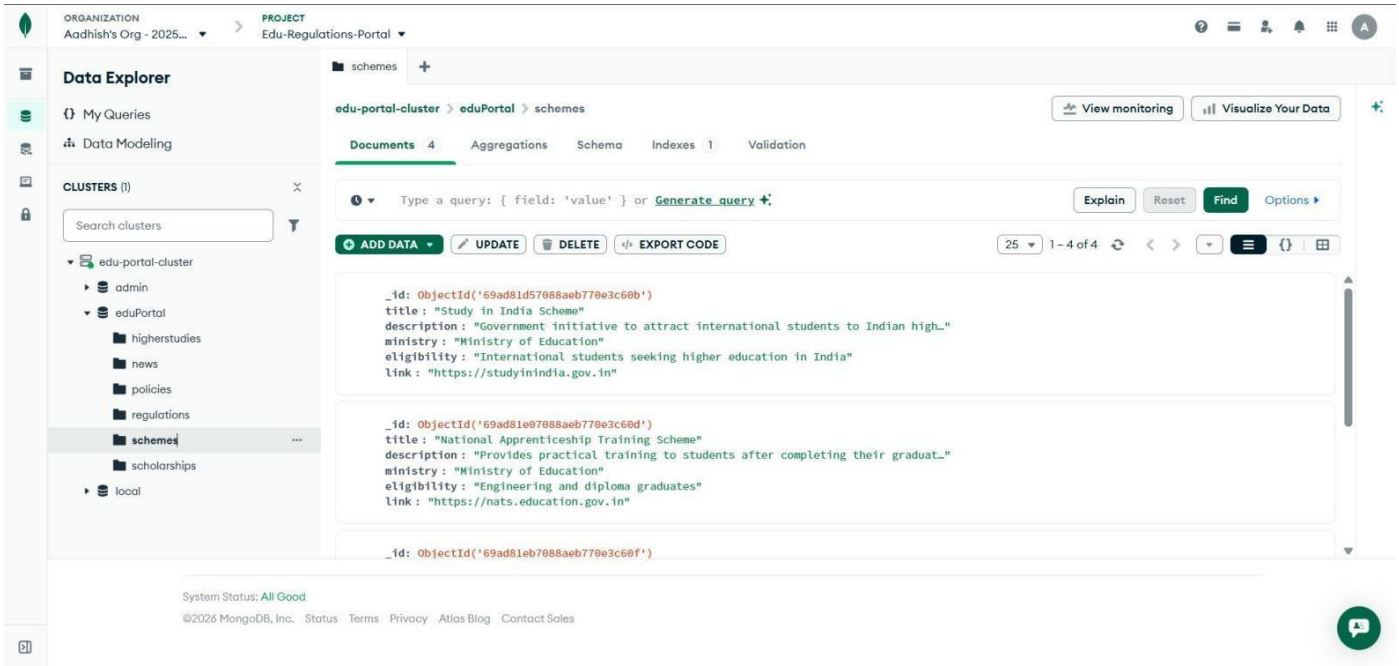
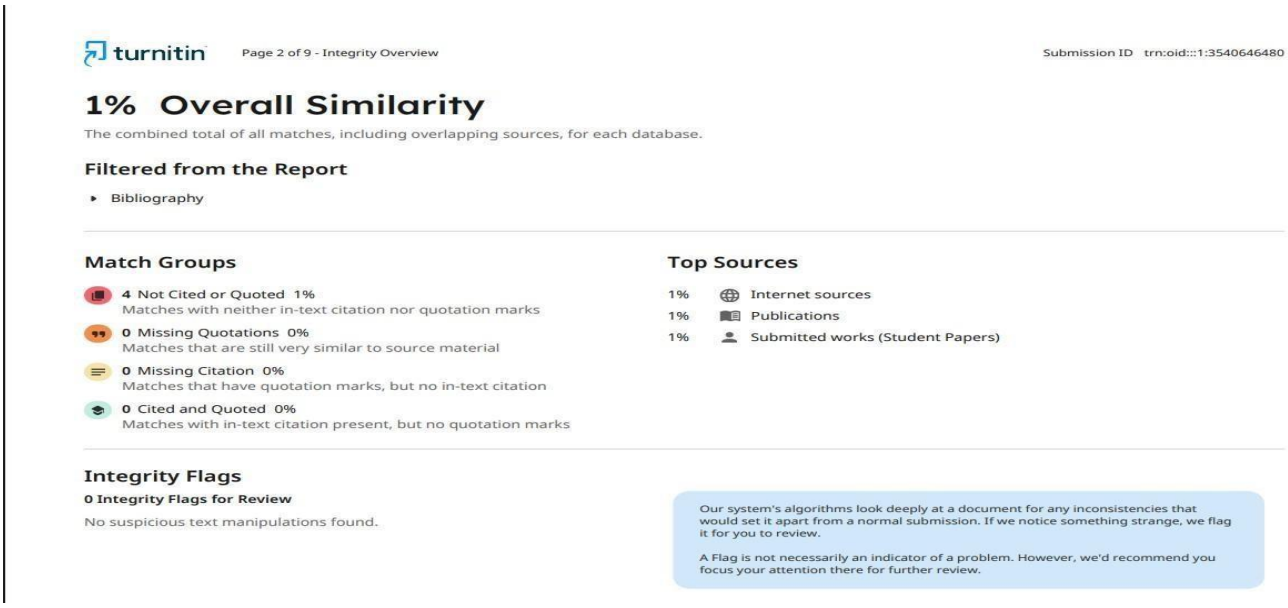


Fig M. MongoDB Atlas – Schemes Collection Data

Appendix – C

Project Similarity Report's



turnitin Page 2 of 9 - Integrity Overview Submission ID trn:oid:::1:3540646480

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Match Groups

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  Matches that are still very similar to source material
- 0 Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

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1% Publications
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Integrity Flags

0 Integrity Flags for Review

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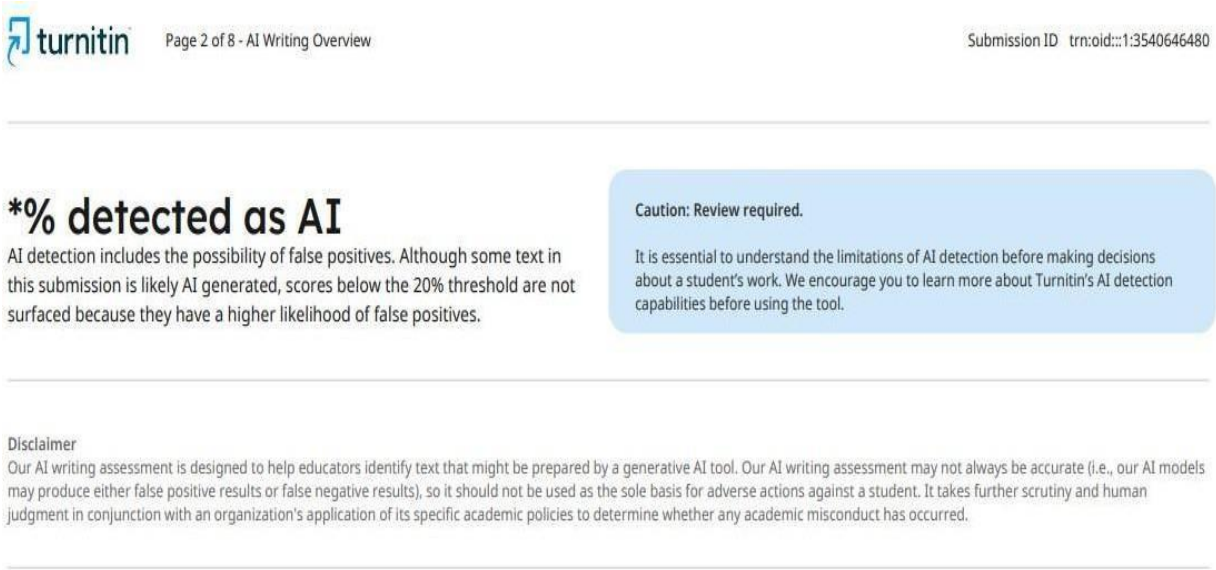
Fig N. Similarity Report



turnitin Page 2 of 8 - AI Writing Overview Submission ID trn:oid:::1:3540646480

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Fig O. AI Similarity Report

Appendix – D

Conference Paper Submission Confirmation

4/23/26, 4:55 PM Gmail - Paper 1818 summary

Gmail

Amith HS <amithhs0109@gmail.com>

Paper 1818 summary
1 message

Microsoft CMT <noreply@msr-cmt.org> Thu, Apr 23, 2026 at 3:33 PM
To: amithhs0109@gmail.com

Hello.

Here is submission summary.

Track Name: Emerging Education Technologies

Paper ID: 1818

Paper Title: Edu-Info-Portal

Abstract:
Educational information such as government policies, academic regulations, scholarship schemes, higher study opportunities, and examination notifications are often distributed across multiple websites and documents, making it difficult for students and researchers to access relevant information efficiently.
This paper presents Edu Info Portal, a web-based centralized information system designed to simplify the retrieval and organization of educational resources.
The system integrates a modern full-stack architecture consisting of a React.js frontend, Node.js and Express.js backend, and MongoDB Atlas cloud database to store and manage structured educational data. The portal categorizes information into multiple sections such as educational policies, government schemes, regulations, scholarships, higher study opportunities, and educational news.
The platform dynamically retrieves information from the database through RESTful APIs, enabling efficient data access and real-time updates without modifying the website structure. The system demonstrates a scalable and modular architecture that can accommodate new categories of educational information as the platform expands.
The current implementation successfully integrates frontend, backend, and cloud database components to create a fully functional centralized educational information portal. Future enhancements include advanced search capabilities, filtering mechanisms, and intelligent recommendation systems to further improve accessibility and user experience.

Created on: Thu, 23 Apr 2026 07:29:47 GMT

Last Modified: Thu, 23 Apr 2026 08:32:13 GMT

Authors:
- amithhs0109@gmail.com (Primary)

Secondary Subject Areas: Not Entered

Submission Files:
Research_Paper_CSE_149_Review_5.docx (167 Kb, Thu, 23 Apr 2026 07:29:42 GMT)

Submission Questions Response: Not Entered

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Fig P. Submission confirmation email for ICACECS 2026 paper